TUTORIAL 7 - Write a C++ program to implement a String class using a default and parameterized constructor along with destructor, Accept() and Display() functions.

NAME – Arenyu Bahiwal PRN – B25ET1073
BATCH – SY 2 DIV – A2

```cpp
#include <iostream>
#include <cstring>
using namespace std;

class String {
    char* str;
public:
    // Default Constructor
    String() {
        str = new char[1]{'\0'};
    }

    // Parameterized Constructor
    String(const char* val) {
        str = new char[strlen(val) + 1];
        strcpy(str, val);
    }

    // Destructor
    ~String() { delete[] str; }

    // Accept Function
    void Accept() {
        char buf[256];
        cout << "Enter text: ";
        cin.getline(buf, 256);
        delete[] str;
        str = new char[strlen(buf) + 1];
        strcpy(str, buf);
    }

    // Display Function
    void Display() { cout << str << endl; }
};

int main() {
    String s1;          // Default
    s1.Accept();
    s1.Display();

    String s2("Hello"); // Parameterized
    s2.Display();
    return 0;
}
```

Output

```
Enter text: arenyu
arenyu
Hello
```